Edge cases (when the answer spans two chunks) — Solution

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Story — `chunking_practice.py` (Edge cases section): you build this exact failure on purpose, here, where nothing's at stake — so the first time it happens for real, in the Build Task's knowledge base, you recognize it immediately instead of trusting a confident, half-right answer. **If not:** "the model gave a plausible but incomplete answer" would look identical to "the model doesn't know," and you'd have no idea which one you were debugging.

Solution

```
# chunking_practice.py — Edge cases section
import math
from openai import OpenAI

client = OpenAI()

DOCUMENT: str = (
    "The quarterly meeting will be held at 3 PM on Friday in Room 204, "
    "and everyone from the product and engineering teams is expected "
    "to attend without exception."
)
CHUNK_SIZE: int = 80
QUESTION: str = (
    "What time and room is the quarterly meeting, and who is expected "
    "to attend?"
)

def chunk_text(text: str, chunk_size: int) -> list[str]:
    results = []
    for i in range(0, len(text), chunk_size):
        results.append(text[i:i + chunk_size])
    return results

def get_embedding(text: str) -> list[float]:
    response = client.embeddings.create(
        model="text-embedding-3-small", input=text
```

```
    )
    return response.data[0].embedding

def cosine_similarity(a: list[float], b: list[float]) -> float:
    dot = 0.0
    for x, y in zip(a, b):
        dot = dot + x * y
    squares = 0.0
    for x in a:
        squares = squares + x * x
    norm_a = math.sqrt(squares)
    squares = 0.0
    for x in b:
        squares = squares + x * x
    norm_b = math.sqrt(squares)
    return dot / (norm_a * norm_b)

def search(
    question: str,
    chunks: list[str],
    chunk_embeddings: list[list[float]],
    k: int,
) -> list[str]:
    question_embedding = get_embedding(question)
    scored = []
    for chunk, embedding in zip(chunks, chunk_embeddings):
        score = cosine_similarity(question_embedding, embedding)
        scored.append((score, chunk))
    # how: tuples sort by their first item — the score — highest first
    scored.sort(reverse=True)
    top_chunks = []
    for score, chunk in scored[:k]:
        top_chunks.append(chunk)
    return top_chunks

# Cut the document into fixed-size chunks and see where the facts landed.
chunks = chunk_text(DOCUMENT, CHUNK_SIZE)
print(f"document was cut into {len(chunks)} chunks:")
for i, chunk in enumerate(chunks):
    print(f"  {i}: {chunk!r}")

# Embed each chunk once.
chunk_embeddings = []
for chunk in chunks:
    chunk_embeddings.append(get_embedding(chunk))

# Compare retrieving just the top chunk (k=1) vs all of them.
```

```
top_1 = search(QUESTION, chunks, chunk_embeddings, k=1)
top_all = search(QUESTION, chunks, chunk_embeddings, k=len(chunks))

print("\nk=1 result:")
for chunk in top_1:
    print(" ", chunk)
print(f"\nk={len(chunks)} result (all chunks):")
for chunk in top_all:
    print(" ", chunk)
```